

SIDDHARTH BHOPLE

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🐙 github.com/SiD-array

Education

Rochester Institute of Technology

Master of Science in Computer Science, GPA: 3.82

Aug. 2024 – May 2027

Rochester, NY

Institute of Engineering and Science, IPS Academy

Bachelor of Technology in Computer Science

Dec. 2020 – Jun. 2024

Indore, India

Experience

BSH Hausgeräte GmbH (Bosch Group)

May 2026 – Present

DAAD RISE Professional Scholar — AI Engineering Intern (Pre-Development)

Berlin, Germany

- Engineered filtering and merge logic processing 1,000+ TDMS files (avg. 1.5GB each) across multi-campaign lab experiments, validating model performance to within +/-500g RMSE target
- Automated the TDMS processing pipeline in Python (Pandas, openpyxl), replacing 2 manual MATLAB workflows and cutting processing time by ~75% (2 hours to 30 minutes)
- Fine-tuned a MobileNetV2 model in TensorFlow to classify 25 fabric types into 5 functional wash categories, improving validation accuracy from 66.4% to 71.5% via early-stopping optimization
- Developing regression and PCA-based feature reduction pipeline across 4+ sensor channels to support Remaining Useful Life (RUL) modeling for embedded laundry appliance diagnostics

Verzeo EduTech

May 2021 – Jul. 2021

Machine Learning Intern

Indore, India

- Implemented ML classification models (Naive Bayes, SVM, Logistic Regression) with 71% accuracy — 15% above the untrained baseline — on a real-world text classification dataset
- Streamlined preprocessing pipeline, reducing sparse features by 30% to improve model training efficiency
- Created a content-based movie recommendation system processing 5,000+ titles and delivering top-5 personalized results in under 1 second per query
- Developed interactive dashboards to analyze user behavior and feature correlations, reducing reporting effort by 50%

Projects

Autonomous Cloud Governance | *Python, AWS, React, Multi-Agent Systems*

Mar. 2026

- Designed a cloud-native multi-agent governance platform integrating 4+ autonomous AI agents for cost-aware orchestration, intelligent routing, and policy-driven decision-making across distributed workflows
- Built a fiscal governance system with circuit breakers and cognitive-tiered routing, cutting cost overrun from 412% to 0% while maintaining full task completion (80% reduction on routine workflows)
- Maintained 100% enforcement accuracy across 50+ autonomous iterations with zero race conditions, at sub-millisecond governance latency (0.04ms, <0.01% overhead vs. LLM inference)
- Architected a modular cloud-native framework utilizing Python, React, and Amazon Web Services to scale governance automation for over 100 concurrent agents with zero performance bottlenecks

RNA-Seq Biomarker Discovery | *Python, Scikit-learn, Pandas, NumPy, gProfiler*

Dec. 2025

- Developed an interpretable ML pipeline in Python (scikit-learn) for multi-class cancer subtype classification across 5 TCGA RNA-Seq datasets, applying cross-validation, regularization, and pathway enrichment analysis
- Attained 100.0% mean 5-fold cross-validated accuracy (std dev 0.0%) with regularized Logistic Regression, outperforming SVC (99.88%) and Random Forest (99.63%) baselines
- Reduced 1,000+ gene features to a 50-gene biomarker panel via coefficient-based feature selection, validated through g:Profiler pathway enrichment analysis

Real-Time Stock Analysis Dashboard | *React (Vite), Recharts, Firebase, FastAPI*

Jul. 2025

- Deployed a full-stack stock analytics platform using React (Vite), FastAPI, and Firebase with real-time market data pipelines and 5+ technical indicators for multi-horizon trend analysis
- Backtested Random Forest across 5 U.S. stocks (AAPL, MSFT, TSLA, GOOGL, AMZN) using 80/20 train-test split, delivering 5.13% average MAPE across all tickers
- Evaluated 4 forecasting models (Random Forest, XGBoost, LightGBM, Linear Regression) on live AAPL data; Linear Regression achieved lowest error (MAPE 4.24%, MAE 12.15)

Technical Skills

Languages: Python, Java, JavaScript, C/C++, C#/.NET, SQL, HTML/CSS

AI/ML: Scikit-learn, TensorFlow, PyTorch, XGBoost, NumPy, Pandas, Feature Engineering, PCA, RUL Modeling

Frameworks & Tools: React, Node.js, FastAPI, Streamlit, Vite, Git, GitHub, VS Code, MATLAB

Cloud & Databases: AWS, Firebase, MySQL, PostgreSQL, MongoDB